

Adopted Levels, Gammas

$Q(\beta^-)=167.322\ 26$; $S(n)=6985.84\ 4$; $S(p)=11586.5\ 8$; $Q(\alpha)=-8322.09\ 6$ [2021Wa16](#)

$S(2n)=18402.99\ 4$, $S(2p)=22909.8\ 7$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Cl}(n,X)$ reaction at Physical Institute of Aarhus University ([1936An01](#)).

^{35}S decay measurements:

[2014Si21](#), [2000Ho13](#), [1999Pa18](#), [1999Oh09](#).

^{35}S cross-section measurements:

$^{35}\text{Cl}(n,p)^{35}\text{S}$: [2024Ha18](#), [2023De34](#), [2021WaZY](#), [2020Ku17](#), [2019Ba16](#), [2000Ka30](#), [1999Gl09](#), [1999Gl02](#);

Projectile fragmentation $^9\text{Be}(^{40}\text{Ar},X)$ and $^{181}\text{Ta}(^{40}\text{Ar},X)$: [2012Zh06](#), [2007No13](#);

Spallation $^{136}\text{Xe}(p,X)$: [2007Na31](#);

$^{34}\text{S}(n,\gamma)$ E=thermal: [2023KoZS](#), [2008FiZZ](#).

 ^{35}S LevelsCross Reference (XREF) Flags

A	$^{35}\text{P}\ \beta^-$ decay (47.3 s)	F	$^{34}\text{S}(n,\gamma)$ E=thermal	K	$^{37}\text{Cl}(p,^3\text{He})$
B	$^2\text{H}(^{34}\text{S},p\gamma)$	G	$^{34}\text{S}(n,n),(n,\gamma)$:resonances	L	$^{37}\text{Cl}(d,\alpha\gamma)$
C	$^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$	H	$^{34}\text{S}(\text{pol } d,p)$	M	$^{160}\text{Gd}(^{34}\text{S},x\gamma),(^{37}\text{Cl},X\gamma)$
D	$^{24}\text{Mg}(^{14}\text{N},3p\gamma)$	I	$^{34}\text{S}(d,p)$	N	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$
E	$^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$	J	$^{34}\text{S}(d,p\gamma)$		

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
0.0	3/2 ⁺	87.61 d 24	ABCDEF HIJKLMN	$\% \beta^- = 100$ $\mu = (+)1.00\ 4$ (1954Bu05 , 2019StZV) $Q = 0.0483\ 3$ (2018Py01 , 2021StZZ) μ : +1.00 4 or -1.07 4 from 1954Bu05 using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes. Q : from 2018Py01 using the microwave absorption method. -0.483 3 is given in 2021StZZ . Other: 0.045 from 1954Bi40 using the microwave absorption method. J^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L-1/2 transfer from analyzing power. J=3/2 also from microwave spectroscopy (1949Co12 , 1951We11). $T_{1/2}$: unweighted average of 88 d 3 (1941Ka01), 87.1 d 12 (1947Hu06), 86.35 d 17 (1959Co56), 87.16 d 10 (1958Se49), 88.8 d 10 (1959Ca12), 89.0 d 5 (1961Wy01), 87.1 d 9 (1961Oz01), 87.9 d 3 (1965Fl02), 87.39 d 10 (1968Wo06), 87.5 d 4 (1969La34), and 87.38 d 3 (1999Pa18).
1572.370 10	1/2 ⁺	2.29 ps 14	ABCD F HIJKL N	J^π : L=0 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$. $T_{1/2}$: from lifetime=3.3 ps 2 in $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$ from 2022Gr07 with DRDM. Others: lifetime=3.3 ps 5 in ($^{34}\text{S},p\gamma$) from 1973Wa10 with DSAM; lifetime>1.8 ps in (d,p γ) from 1970Bu18 with DSAM; lifetime>1.6 ps in (d,p γ) from 1972Fr11 with DSAM.
1991.27 4	7/2 ⁻	1.03 ns 5	CDEF HIJKLMN	J^π : L(pol d,p)=3 from 0 ⁺ and L+1/2 transfer from analyzing power. $T_{1/2}$: from lifetime=1.48 ns 7: weighted average of 1.7 ns 3 from (d,p γ) from 2024Co04 with p γ -delayed coin and 1.47 ns 7 from (d,p γ) from 1971Pr11 with p γ -delayed coin. Other: >4.5 ps in (d,p γ) (1970Bu18 , DSAM).
2347.779 10	3/2 ⁻	0.78 ps 12	BCD F HIJ L N	J^π : L=1 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2 transfer from analyzing power. $T_{1/2}$: from lifetime=1.12 ps 18: weighted average of 1.3 ps 2 in ($^{34}\text{S},p\gamma$) from 1973Wa10 with DSAM, 0.8 ps 2 in (d,p γ) from 1970Bu18 with DSAM, and 1.22 ps 18 in (d,p γ) from 1972Fr11 with DSAM.

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
2717.07 9	5/2 ⁺	69 fs 24	CD F HIJKL N	J ^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime=100 fs 35 in (d,pγ) from 1972Fr11 with DSAM.
2938.62 5	(3/2) ⁺		A C F HIJKL	J ^π : L=2 from 0 ⁺ in (d,p) and (pol d,p); J=3/2, 5/2 from γ(θ) in (d,pγ) (1972Va07,1975VaYG); Possibly allowed β ⁻ feeding from 1/2 ⁺ parent with log ft=5.0. Other: L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ is inconsistent.
3423.2 20	5/2 ⁺	<69 fs	C HIJKL N	J ^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime<100 fs in (d,pγ) from 1972Fr11 with DSAM.
3558.079 28	(3/2 ⁻ ,5/2 ⁺)		F HIJ L N	J ^π : 1986γ to 1572, 1/2 ⁺ and 1566.7γ to 1991, 7/2 ⁻ .
3594.62 24	(7/2 ⁺)		CDE IJKL N	XREF: C(3602)I(3596)K(3598) J ^π : 3594.5γ Q, ΔJ=2 to g.s., 3/2 ⁺ . Other: L=2 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ is inconsistent.
3675? 10	(1/2 ⁻ ,3/2 ⁻)		I	J ^π : L=(1) from 0 ⁺ in (d,p).
3801.952 16	3/2 ⁻	25 fs 18	F HIJKL	XREF: k(3811) J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime=36 fs 26 in (d,pγ) from 1972Fr11 with DSAM.
3816.01 21	(9/2 ⁻)	0.28 ps 3	CDE H JKL N	XREF: k(3811) J ^π : 207.0γ from 4023, (11/2 ⁻). 2061.6γ D, ΔJ=1 from 5877.7 (11/2 ⁺). T _{1/2} : from ($^{14}\text{N},3\text{p}\gamma$) in 2014Ay01 with DSAM.
3886.28 33	(5/2 ⁺)		CD IJ L	XREF: C(3895) J ^π : 1894.9γ D, ΔJ=1 to 1991, 7/2 ⁻ . 3886γ to g.s., 3/2 ⁺ . Other: L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$ is inconsistent.
4023.24 21	(11/2 ⁻)	0.32 ps 3	DE LMN	J ^π : 2032.0γ E2, ΔJ=2 to 1991, 7/2 ⁻ . T _{1/2} : from ($^{14}\text{N},3\text{p}\gamma$) in 2014Ay01 with DSAM.
4027.2 14	3/2 ⁺ ,5/2 ⁺		C IJKL	XREF: I(4025) J ^π : L=2 from 0 ⁺ in $^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$.
4105.6 8	(3/2 ⁻ ,5/2 ⁺)	<55 fs	F IJKL	J ^π : 2533γ to 1572, 1/2 ⁺ and 2114γ to 1991, 7/2 ⁻ . T _{1/2} : from lifetime<80 fs in (d,pγ) from 1972Fr11 with DSAM.
4180 3	(1/2,3/2,5/2 ⁺)		kL	XREF: k(4186) J ^π : 2608γ to 1572, 1/2 ⁺ .
4189.238 33	1/2 ⁻	<35 fs	F HIJKL	XREF: k(4186)L(4187) J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power. T _{1/2} : from lifetime<50 fs in (d,pγ) from 1972Fr11 with DSAM.
4302.6 7			I KL	XREF: K(4290)
4477.63 7	(5/2 ⁺)	<62 fs	F iJKL	XREF: i(4481)K(4489) J ^π : 2905γ to 1572, 1/2 ⁺ and 2486γ to 1991, 7/2 ⁻ . L=2 from 3/2 ⁺ in (p, ^3He). L=2 component of L=2,3 doublet from 0 ⁺ in (d,p). T _{1/2} : from lifetime<90 fs in (d,pγ) from 1972Fr11 with DSAM.
4482	7/2 ⁻		Hi	XREF: i(4481) J ^π : L=3 from 0 ⁺ in (pol d,p) and L+1/2 transfer from analyzing power. L=3 component of L=2,3 doublet from 0 ⁺ in (d,p).
4576 8	(⁺)		HI K	E(level): weighted average of 4575 8 from (d,p) and 4577 10

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
				E(level): weighted average of 4575 8 from (d,p) and 4577 10 from (p, ³ He).
				J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He). L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
4617 10			K	
4822.79 24	(9/2 ⁺)		DE	J ^π : 1228.1γ D, ΔJ=1 to 3594.6, (7/2 ⁺). 1055.1γ D, ΔJ=1 from 5877.7, (11/2 ⁺). 1006.8γ D, ΔJ=0 to 3816.0, (9/2 ⁻).
4839 8	(⁺)		HI K	E(level): weighted average of 4837 8 from (d,p) and 4843 10 from (p, ³ He). Other: 4837 from (pol d,p).
				J ^π : L=2 from 3/2 ⁺ in (p, ³ He). L=2 gives 1/2 ⁺ to 9/2 ⁺ .
4899.82 27	(9/2 ⁺)		DE	J ^π : 2908.3γ D, ΔJ=1 to 1991, 7/2 ⁻ . 1055.1γ D, ΔJ=1 from 5877.7, (11/2 ⁺).
4903.371 17	1/2 ⁻		F HI	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power.
4963.081 25	3/2 ⁻		F HI K	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
4990 10	(⁺)		K	J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He). L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
5010.05 29	(11/2 ⁻)	0.45 ps 8	DE	J ^π : 986.8γ D, ΔJ=0 to 4023, (11/2 ⁻) and 3018.5γ E2 to 1991, 7/2 ⁻ . T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
5058 8	7/2 ⁻		HI	E(level): from (d,p).
				J ^π : L=3 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
5127 10	(⁺)		HI K	E(level): from (p, ³ He). Other: 5126 11 from (d,p).
				J ^π : L=2 from 3/2 ⁺ in (p, ³ He). L=2 gives (1/2 ⁺ to 9/2 ⁺).
5343 8	(⁻)		I K	E(level): weighted average of 5342 8 from (d,p) and 5345 10 from (p, ³ He).
				J ^π : L=3 from 3/2 ⁺ in (p, ³ He). L=3 gives (1/2 ⁻ to 11/2 ⁻).
5412.28 25	(9/2 ⁺)		DE	J ^π : 1389.0γ D, ΔJ=1, to 4023, (11/2 ⁻). 3420.6γ to 1991, 7/2 ⁻ . 1525.7γ to 3886, (5/2 ⁺).
5550 10			I K	XREF: I(5542?)
				J ^π : L=(3) from 3/2 ⁺ in (p, ³ He). L=3 gives (1/2 ⁻ to 11/2 ⁻).
5752.5 8	5/2 ⁻ , 7/2 ⁻		F I	XREF: I(5740)
				J ^π : L=3 from 0 ⁺ in (d,p); 5752γ to 3/2 ⁺ .
5771 10	(⁺)		K	J ^π : L=2 from 3/2 ⁺ in (p, ³ He). L=2 gives (1/2 ⁺ to 9/2 ⁺).
5841.482 31			F	
5877.70 22	(11/2 ⁺)		DE	J ^π : 465.4γ D, ΔJ=1 to 5412, (9/2 ⁺). 867.6γ D, ΔJ=0 to 5010, (11/2 ⁻). 3886γ to 1991, 7/2 ⁻ . 2283γ to 3594.6, (7/2 ⁺).
5890 20	3/2 ⁺ , 5/2 ⁺		I K	XREF: K(5915?)
				J ^π : L=2 from 0 ⁺ in (d,p).
5980 10			I	
6018.8 6			F	
6078.475 33	1/2 ⁻ , 3/2 ⁻		F I	J ^π : L=1 from 0 ⁺ in (d,p).
6129 10			K	J ^π : L=0+2 from 3/2 ⁺ in (p, ³ He). L=0 gives (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺); L=2 gives (1/2 ⁺ to 9/2 ⁺).
6293.93 6			F I	XREF: I(6292)
				E(level): other: 6292 8 from (d,p).
6299.4 4	(11/2 ⁺)		D	J ^π : 887.0γ, ΔJ=1 to 5412, (9/2 ⁺). 2275.9γ to 4023, (11/2 ⁻).
6334 8			I	
6344 8			I K	XREF: K(6347)
6352.2 5	(13/2 ⁻)	50 fs 10	D	J ^π : 2329.5γ D, ΔJ=1 to 4023, (11/2 ⁻). 2535.8γ E2 to 3816.0, (9/2 ⁻).
				T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
6354.89 23			F	

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
6419.9 11			F	
6446 8			I	
6496 8			I	
6537.7 14			I	
6545.1 13			I	
6584 10			I	
6629.43 9			F	
6635.2 13			I	
6654 10			K	
6684 9			I K	E(level): weighted average of 6677 8 from (d,p) and 6696 10 from (p, ³ He). J ^π : L=2 from 3/2 ⁺ in (p, ³ He). L=2 gives (1/2 ⁺ to 9/2 ⁺).
6761.0 12			F	
6891.3 14	(⁻)		I	J ^π : L=(1,3) from 0 ⁺ in (d,p).
(6986.096 25)	1/2 ⁺		F	J ^π : s-wave resonance capture state.
7018.89 4			G I	E(level): from (n,n). Other: 7021 10 from (d,p).
7072.66 5			G	
7074.33 6			G	
7097.72 30			G	
7099.85 20			G	
7100.73 4	3/2 ⁻		G	
7143.27 11			G I K	XREF: I(7150)K(7151)
7179.74 32	(15/2 ⁺)	3.1 ps 12	DE	J ^π : 1302.0γ E2 to 5877.7, (11/2 ⁺); γ to 6352.2, (13/2 ⁻). T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM. XREF: I(7205)
7210.42 5			G I	
7218.0 4			G	
7234.0 5			G	
7239.8 6			G	
7253.2 6			G	
7275.93 5	1/2 ⁺		G	
7289.8 8			G	
7294.19 6	3/2 ⁻		G	
7331.00 5	1/2 ⁺		G	
7338.18 20			G	
7343.91 20			G	
7347.2 5			G	
7347.9 5			G	
7353.9 5			G	
7357.6 5			G	
7370.56 6	1/2 ⁻		G K	XREF: K(7375)
7396.0 6			G	
7404.6 6			G	
7408.68 11	3/2 ⁻		G	
7411.55 6			G	
7416.50 4			G	
7429.1 6			G	
7433.56 5			G	
7436.4 4			G	
7442.14 5	1/2 ⁺		G	
7462.2 6			G	
7481.15 7	1/2 ⁻		G I	E(level): weighted average of 7481.15 6 from (n,n) and 7482.7 13 from (d,p).
7494.5 6			G	
7542.3 6			G i	XREF: i(7570)
7604.0 6			G i	XREF: i(7570)
7609.26 5	3/2 ⁻		G	

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
7648.9 6			G i	XREF: i(7640)
7655.7 6			G i	XREF: i(7640)
7663.91 11			G	
7678.3 7			G	
7731.1 7			G K	XREF: K(7712?)
7749.7 4	(3/2)		G	
7761.46 7	3/2 ⁻		G k	XREF: k(7770)
7776.17 11	1/2 ⁻		G k	XREF: k(7770)
7797.99 9	1/2 ⁺		G	
7812.24 7	3/2 ⁺		G	
7853.25 7	3/2 ⁻		G	
7861.8 12			G	
7880.8 12			G	
7889.0 12			G	
7894.63 7	3/2 ⁻		G	
7899.70 30			G	
7934.1 15			G	
7939.5 15			G	
7954.96 11	3/2 ⁻		G	
7974.19 11	3/2 ⁻		G	
8019.5 2			G	
8024.0 7	(17/2 ⁺)	0.15 ps 4	DE	J ^π : 844.2γ D, ΔJ=1 to 7179.7, (15/2 ⁺). T _{1/2} : from (^{14}N ,3pγ) in 2014Ay01 with DSAM.
8041.20 20			G	
8076.46 20			G	
8077.91 30			G	
8093.0 5	1/2 ⁻		G K	XREF: K(8103) E(level): from (n,n). Other: 8103 10 from (p, ^3He).
8143.75 20			G	
8160 10			K	
8187.17 30	1/2 ⁻		G	
8211.25 30			G	
8224.2 4	1/2 ⁻		G	
8228.3 4			G	
8243.98 30			G	
8256.31 20	3/2 ⁻		G	
8262.4 4	5/2 ⁺		G	
8273.3 4	(1/2 ⁻)		G	
8298.27 30			G	
8333.8 4	3/2 ⁻		G	
8335.9 4			G	
8391.3 4	3/2 ⁻		G	
8393.64 30			G	
8406.55 30			G k	XREF: k(8430)
8418.8 4			G k	XREF: k(8430)
8755.9 11	(19/2)	<1 ps	E	J ^π : L=2 from 3/2 ⁺ in (p, ^3He). L=2 gives (1/2 ⁺ to 9/2 ⁺). J ^π : 731.6γ D, ΔJ=1 to (17/2 ⁺). T _{1/2} : from (^{18}O ,2αnγ) in 2021Go09 with residual Doppler energy shifts observed for the 732γ and 1576γ peaks.
9155 10			K	J ^π : L=2 from 3/2 ⁺ in (p, ^3He). L=2 gives (1/2 ⁺ to 9/2 ⁺).
10048.6 9	(19/2)		E	J ^π : 2868.7γ Q, ΔJ=2 to (15/2 ⁺).
12469.3 14	(21/2)		E	J ^π : 2420.6γ D, ΔJ=1 to (19/2).

[†] From a least-squares fit to γ-ray energies with uncertainties for levels connected with γ transitions; from $^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances for resonances.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	$\alpha^@$	Comments
1572.370	1/2 ⁺	1572.327 12	100	0.0	3/2 ⁺	(M1)		0.0001007 14	B(M1)(W.u.)=0.00248 15 E _γ : weighted average of 1572.252 29 from ³⁵ P β ⁻ decay, 1576 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 1572.4 3 from (¹⁴ N,3pγ), 1572.333 8 from (n,γ) E=thermal, 1572.24 17 from (d,pγ), and 1572 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). Mult.: D from (¹⁴ N,3pγ); Δπ=no from level scheme.
1991.27	7/2 ⁻	1991.26 5	100	0.0	3/2 ⁺	M2+E3	+0.17 6	0.0001476 22	B(M2)(W.u.)=0.087 5; B(E3)(W.u.)=3.7 +28-21 E _γ : weighted average of 1990 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 1991.3 2 from (¹⁴ N,3pγ), 1990.5 4 from (¹⁸ O,2αnγ), 1991.27 5 from (n,γ) E=thermal, and 1991 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). Mult.,δ: from (d,pγ). D+Q or Q+O from γ(θ) in (d,pγ); E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2 ⁻ to 3/2 ⁺ .
2347.779	3/2 ⁻	356.66& 9 775.398 6	<0.084& 35.3 27	1991.27 7/2 ⁻ 1572.370 1/2 ⁺	[E2] (E1)			1.15×10 ⁻³ 2 3.54×10 ⁻⁵ 5	E _γ ,I _γ : from (n,γ) E=thermal. B(E1)(W.u.)=4.5×10 ⁻⁴ +9-7 E _γ : from (n,γ) E=thermal. Others: 778 2 from ⁹ Be(³⁶ S, ³⁵ Sγ) and 775.4 4 from (¹⁴ N,3pγ). I _γ : weighted average of 37 5 from (¹⁴ N,3pγ), 35 4 from (n,γ) E=thermal, 37.0 28 from (d,pγ), and 33.3 27 from (d,αγ). Mult.: D(+Q) from γ(θ) in (d,pγ); Δπ=yes from level scheme; M2 ruled out by RUL.
		2347.700 19	100.0 27	0.0	3/2 ⁺	(E1)		0.000880 12	B(E1)(W.u.)=4.6×10 ⁻⁵ +9-6 E _γ : weighted average of 2352 2 from ⁹ Be(³⁶ S, ³⁵ Sγ), 2347.8 3 from (¹⁴ N,3pγ), 2347.700 15 from (n,γ) E=thermal, and 2347 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from (d,αγ). Others: 100 8 from (¹⁴ N,3pγ), 100 8 from (n,γ) E=thermal, and 100.0 28 from (d,pγ). Mult.: D(+Q) from γ(θ) in (d,pγ); Δπ=yes from level scheme; M2 ruled out by RUL.
2717.07	5/2 ⁺	368.7 5	14 8	2347.779 3/2 ⁻	[E1]			0.0002294 33	B(E1)(W.u.)=0.022 +19-10 E _γ : weighted average of 368.5 4 from (n,γ) E=thermal and 370 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : unweighted average of 6.7 23 from (n,γ) E=thermal and 22 4 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
		725.79 1144.68 2717.01 16	<5 <5 100 10	1991.27 7/2 ⁻ 1572.370 1/2 ⁺ 0.0 3/2 ⁺	[E1] [E2] M1+E2			4.10×10 ⁻⁵ 6 3.79×10 ⁻⁵ 5 0.000568 9	I _γ : from (d,pγ). I _γ : from (d,pγ). B(M1)(W.u.)=0.012 +7-3; B(E2)(W.u.)=0.9 +6-3

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\alpha^@$	Comments
								B(M1)(W.u.)=0.012 +7-3; B(E2)(W.u.)=0.9 +6-3 E_γ : weighted average of 2720 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, 2717.0 4 from ($^{14}\text{N}, 3\text{py}$), 2716.99 16 from (n, γ) E=thermal, and 2717 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 100 17 from (n, γ) E=thermal. Mult., δ : D+Q from $\gamma(\theta)$ in (d,py) with $\delta=-0.39$ 5 or -15.0 34 for $J_i=5/2$ (1975VaYG); -15.0 34 is unlikely based on RUL; E1+M2 is ruled out by RUL.
2938.62	(3/2) ⁺	590.84	<10	2347.779	3/2 ⁻			I_γ : from (d,py).
		947.34	<20	1991.27	7/2 ⁻			I_γ : from (d,py).
		1366.22	<15	1572.370	1/2 ⁺			I_γ : from (d,py).
		2938.57 18	100	0.0	3/2 ⁺			E_γ : weighted average of 2938.29 40 from $^{35}\text{P} \beta^-$ decay, 2943 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, and 2938.58 11 from (n, γ) E=thermal.
3423.2	5/2 ⁺	484.6	<4	2938.62	(3/2) ⁺	[M1,E2]	2.7×10^{-4} 13	I_γ : from (d,py).
		706.1	<6	2717.07	5/2 ⁺	[M1,E2]	9.5×10^{-5} 28	I_γ : from (d,py).
		1075.4	<7	2347.779	3/2 ⁻	[E1]	1.830×10^{-5} 26	I_γ : from (d,py).
		1431.9	<30	1991.27	7/2 ⁻	[E1]	0.0002257 32	I_γ : from (d,py).
		1850.8	<7	1572.370	1/2 ⁺	[E2]	0.000255 4	I_γ : from (d,py).
		3423 2	100	0.0	3/2 ⁺	[M1,E2]	0.00090 7	E_γ : weighted average of 3425 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ and 3421 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
3558.079	(3/2 ⁻ , 5/2 ⁺)	619.23 19	8.9 17	2938.62	(3/2) ⁺			I_γ : from (d,py).
		1210.28 4	54 6	2347.779	3/2 ⁻			E_γ, I_γ : from (n, γ) E=thermal.
								E_γ : from (n, γ) E=thermal. Other: 1211 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		1566.7 3	100 6	1991.27	7/2 ⁻			I_γ : weighted average of 54 6 from (n, γ) E=thermal and 54 6 from (d, $\alpha\gamma$). Other: 18 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
								E_γ : from (n, γ) E=thermal. Other: 1567 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
								I_γ : from (d, $\alpha\gamma$). Others: 100 19 from (n, γ) E=thermal and 100 10 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		1986 ^a 2	39 6	1572.370	1/2 ⁺			E_γ, I_γ : from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		3558.1 5	18 4	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
3594.62	(7/2 ⁺)	655.99	<10	2938.62	(3/2) ⁺			I_γ : from (d,py).
		877.6 5	9.3 23	2717.07	5/2 ⁺			E_γ : from ($^{14}\text{N}, 3\text{py}$).
								I_γ : from ($^{14}\text{N}, 3\text{py}$). Other: <11 from (d,py).
		1246.82	<15	2347.779	3/2 ⁻			I_γ : from (d,py).
		1603.31	<17	1991.27	7/2 ⁻			I_γ : from (d,py).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	$\alpha^@$	
3594.62	(7/2 ⁺)	3594.5 7	100 9	0.0	3/2 ⁺	Q			E_γ : weighted average of 3594.7 7 from ($^{14}\text{N}, 3\text{p}\gamma$), 3594.4 11 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$), and 3593 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : from ($^{14}\text{N}, 3\text{p}\gamma$). Other: 3602 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ($^{14}\text{N}, 3\text{p}\gamma$). B(E1)(W.u.)= $2.4 \times 10^{-4} + 30 - 12$ E_γ : from (n, γ) E=thermal. I_γ : from (n, γ) E=thermal. Other: <8.9 from (d, $\text{p}\gamma$). B(E1)(W.u.)= $2.0 \times 10^{-4} + 25 - 10$ E_γ : from (n, γ) E=thermal. I_γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from (d, $\text{p}\gamma$). E_γ : from (n, γ) E=thermal. I_γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from (d, $\text{p}\gamma$). B(M1)(W.u.)= $0.024 + 29 - 11$ if M1, B(E2)(W.u.)= $4 \times 10^1 + 5 - 2$ if E2. I_γ : from (d, $\text{p}\gamma$). B(E1)(W.u.)= $9 \times 10^{-4} + 11 - 4$ E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from (d, $\text{p}\gamma$), and 61 5 from (d, $\alpha\gamma$). B(E1)(W.u.)= $2.2 \times 10^{-4} + 26 - 9$ E_γ : from (n, γ) E=thermal. I_γ : from (d, $\alpha\gamma$). Other: 100 9 from (d, $\text{p}\gamma$), 100 9 from (n, γ) E=thermal. B(M1)(W.u.)= $0.0099 + 14 - 12$; B(E2)(W.u.)= $3.4 + 10 - 9$ E_γ : weighted average of 1824.6 2 from ($^{14}\text{N}, 3\text{p}\gamma$), 1824.4 6 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$), and 1827 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 1830 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult., δ : from $\gamma(\theta)$, $\gamma\gamma(\theta)(\text{ADO})$, and RUL in ($^{14}\text{N}, 3\text{p}\gamma$). I_γ : from (d, $\text{p}\gamma$). I_γ : from (d, $\text{p}\gamma$). I_γ : from (d, $\text{p}\gamma$). E_γ : from ($^{14}\text{N}, 3\text{p}\gamma$). Other: 1542 2 from
3801.952	3/2 ⁻	863.28 28	1.3 4	2938.62	(3/2) ⁺	[E1]		2.82×10^{-5} 4	
		1084.79 15	2.2 7	2717.07	5/2 ⁺	[E1]		1.801×10^{-5} 25	
		1454.09 4	18 4	2347.779	3/2 ⁻	[M1,E2]		8.0×10^{-5} 12	
		1810.63	<8.9	1991.27	7/2 ⁻	[E2]		0.0002364 33	
		2229.510 16	88 15	1572.370	1/2 ⁺	[E1]		0.000804 11	
		3801.73 4	100 5	0.0	3/2 ⁺	[E1]		1.60×10^{-3} 2	
3816.01	(9/2 ⁻)	1824.7 3	100	1991.27	7/2 ⁻	M1+E2	+0.55 9	0.000200 4	
3886.28	(5/2 ⁺)	463.1	<30	3423.2	5/2 ⁺				
		947.65	<12	2938.62	(3/2) ⁺				
		1169.19	<15	2717.07	5/2 ⁺				
		1538.4 8	57 33	2347.779	3/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult. #</u>	<u>$\alpha^@$</u>	<u>Comments</u>
3886.28	(5/2 ⁺)	1894.9 4	100 11	1991.27	7/2 ⁻	D		E _{γ} : from (¹⁴ N,3p γ). Other: 1542 2 from ⁹ Be(³⁶ S, ³⁵ S γ). I _{γ} : unweighted average of 24 10 from (¹⁴ N,3p γ) and 89 9 from (d, $\alpha\gamma$). Other: <18 from (d,p γ). E _{γ} ,Mult.: from (¹⁴ N,3p γ). I _{γ} : from (d, $\alpha\gamma$). Other: 100 14 from (¹⁴ N,3p γ) and 100 from (d,p γ). I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). Other: <16 from (d,p γ). B(M1)(W.u.)=0.046 +25-21 if M1. B(E2)(W.u.)=4.0 $\times$ 10 ³ +22-19 exceeds RUL=100 if E2. B(E2)(W.u.)=7.5 +8-7
4023.24	(11/2 ⁻)	2313.83 3886.05 207.0 5	<16 33 7 0.59 30	1572.370 0.0 3816.01	1/2 ⁺ 3/2 ⁺ (9/2 ⁻)	[M1,E2]	0.005 4	E _{γ} : weighted average of 2031.8 3 from (¹⁴ N,3p γ) and 2032.4 4 from (¹⁸ O,2 α n γ). Other: 2032 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ S γ). I _{γ} : from (¹⁴ N,3p γ). Mult.: from $\gamma\gamma(\theta)$ (ADO) and RUL in (¹⁴ N,3p γ). I _{γ} : from (d,p γ). I _{γ} : from (d,p γ). I _{γ} : weighted average of 86 22 from (d,p γ) and 97 12 from (d, $\alpha\gamma$). E _{γ} : from ⁹ Be(³⁶ S, ³⁵ S γ). I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). Other: 92 33 from (d,p γ). E _{γ} : from ⁹ Be(³⁶ S, ³⁵ S γ). I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). Other: 100 33 from (d,p γ). I _{γ} : from (d,p γ).
4027.2	3/2 ⁺ ,5/2 ⁺	604.0 1088.6 1309 2	<14 <28 95 12	3558.079 3423.2 2938.62	(3/2 ⁻ ,5/2 ⁺) 5/2 ⁺ (3/2 ⁺)			E _{γ} : from ⁹ Be(³⁶ S, ³⁵ S γ). I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). Other: 92 33 from (d,p γ). E _{γ} : from ⁹ Be(³⁶ S, ³⁵ S γ). I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). Other: 100 33 from (d,p γ). I _{γ} : from (d,p γ).
4105.6	(3/2 ⁻ ,5/2 ⁺)	2454.7 4027.0 547.5 682.4 1167.0 1388.5 1757.8 2114.3 2533.1 4105.3 8	100 18 <36 <4.6 <4.6 15 7 <12 <5.8 <8.1 <5.8 100 7	1572.370 0.0 3558.079 3423.2 2938.62 2717.07 2347.779 1991.27 1572.370 0.0	1/2 ⁺ 3/2 ⁺ (3/2 ⁻ ,5/2 ⁺) 5/2 ⁺ (3/2 ⁺) 5/2 ⁺ 3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺			E _{γ} : from (n, γ) E=thermal. I _{γ} : from (d,p γ). I _{γ} : from (d, $\alpha\gamma$). I _{γ} : from (d, $\alpha\gamma$). E _{γ} : from (n, γ) E=thermal. I _{γ} : from (n, γ) E=thermal. Other: <7.7 from (d,p γ).
4180	(1/2,3/2,5/2 ⁺)	2608 4180	22 6 100 6	1572.370 0.0	1/2 ⁺ 3/2 ⁺			
4189.238	1/2 ⁻	631.32 & 24	<2.8 &	3558.079	(3/2 ⁻ ,5/2 ⁺)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\alpha^@$	Comments	
4189.238	$1/2^-$	1250.61 5	8.4 10	2938.62	$(3/2)^+$	[E1]	0.0001047 15	E_γ : from (n, γ) E=thermal.	
		1840.2 12	91 10	2347.779	$3/2^-$	[M1,E2]	0.000221 29	I_γ : from (n, γ) E=thermal. Other: <9.6 from (d,p γ). E_γ : from (n, γ) E=thermal.	
		2616.8 13	9.5 28	1572.370	$1/2^+$	[E1]	1.04×10^{-3} 2	I_γ : weighted average of 87 16 from (n, γ) E=thermal and 92 10 from (d,p γ). E_γ : from (n, γ) E=thermal.	
		4188.95 5	100 10	0.0	$3/2^+$	[E1]	1.75×10^{-3} 3	I_γ : from (n, γ) E=thermal. Other: <12 from (d,p γ). E_γ : from (n, γ) E=thermal.	
4302.6	$(5/2^+)$	1954.8	100 9	2347.779	$3/2^-$			I_γ : from (d,p γ). Other: 100 11 from (n, γ) E=thermal.	
		4302.3	70 9	0.0	$3/2^+$			I_γ : from (d, $\alpha\gamma$).	
4477.63		591.35	<8.2	3886.28	$(5/2^+)$	[M1,E2]	1.5×10^{-4} 6	I_γ : from (d,p γ).	
		675.67	<8.2	3801.952	$3/2^-$	[E1]	4.82×10^{-5} 7	I_γ : from (d,p γ).	
		883.00	<8.2	3594.62	$(7/2^+)$	[M1,E2]	5.4×10^{-5} 12	I_γ : from (d,p γ).	
		919.54	<8.2	3558.079	$(3/2^-,5/2^+)$			I_γ : from (d,p γ).	
		1054.4	<16	3423.2	$5/2^+$	[M1,E2]	3.6×10^{-5} 6	I_γ : from (d,p γ).	
		1538.97	<12	2938.62	$(3/2)^+$			I_γ : from (d,p γ).	
		1760.55 11	100 12	2717.07	$5/2^+$	[M1,E2]	0.000189 25	E_γ : from (n, γ) E=thermal. I_γ : from (d,p γ). Others: 100 15 from (n, γ) E=thermal and 100 11 from (d, $\alpha\gamma$).	
		2129.78	<13	2347.779	$3/2^-$	[E1]	0.000738 10	I_γ : from (d,p γ).	
2486.27		<16	1991.27	$7/2^-$	[E1]	0.000962 13	I_γ : from (d,p γ). Other: 42 11 from (d, $\alpha\gamma$).		
2905.1 4		90 14	1572.370	$1/2^+$	[E2]	0.000749 10	E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 96 28 from (n, γ) E=thermal, 64 12 from (d,p γ), and 109 11 from (d, $\alpha\gamma$).		
4822.79		$(9/2^+)$	4477.32	<20	0.0	$3/2^+$	[M1,E2]	0.00126 8	I_γ : from (d,p γ). Other: 28 6 from (d, $\alpha\gamma$).
			1006.8 2	66 15	3816.01	$(9/2^-)$	D		E_γ : from (^{14}N ,3p γ). Other: 1007.2 11 from (^{18}O ,2 α n γ). I_γ : weighted average of 73 9 from (^{14}N ,3p γ) and 36 18 from (^{18}O ,2 α n γ).
			1228.1 3	100 9	3594.62	$(7/2^+)$	D		Mult.: from (^{14}N ,3p γ) and (^{18}O ,2 α n γ). E_γ : from (^{14}N ,3p γ). Other: 1228.3 6 from (^{18}O ,2 α n γ). I_γ : from (^{14}N ,3p γ). Other: 100 18 from (^{18}O ,2 α n γ).
			2831.4 9	36 9	1991.27	$7/2^-$			Mult.: from (^{14}N ,3p γ). E_γ, I_γ : from (^{14}N ,3p γ).
4899.82		$(9/2^+)$	1305.3 4	55 18	3594.62	$(7/2^+)$			E_γ : from (^{18}O ,2 α n γ). Other: 1304.8 12 from (^{14}N ,3p γ). I_γ : from (^{14}N ,3p γ).
			2908.3 5	100 18	1991.27	$7/2^-$	D		E_γ, I_γ : from (^{14}N ,3p γ). Mult.: from $\gamma\gamma(\theta)$ (ADO) in (^{14}N ,3p γ).
4903.371		$1/2^-$	1101.92 31	0.43 11	3801.952	$3/2^-$			
			1964.8 2	4.9 13	2938.62	$(3/2)^+$			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\alpha^@$	Comments
4903.371	1/2 ⁻	2555.492 14	41 4	2347.779	3/2 ⁻			
		3330.84 13	100 9	1572.370	1/2 ⁺			
		4902.99 5	53 5	0.0	3/2 ⁺			
4963.081	3/2 ⁻	1161.05 20	0.98 14	3801.952	3/2 ⁻			
		1404.967 24	10.7 11	3558.079	(3/2 ⁻ , 5/2 ⁺)			
		2615.2 2	19.5 18	2347.779	3/2 ⁻			
		2972.0 4	3.2 11	1991.27	7/2 ⁻			
		3390.62 7	100 9	1572.370	1/2 ⁺			
		4963.06 22	47 5	0.0	3/2 ⁺			
5010.05	(11/2 ⁻)	986.8 3	100 5	4023.24	(11/2 ⁻)	D		B(E1)(W.u.)=0.00102 +23-16; B(M1)(W.u.)=0.036 +8-6 E γ : weighted average of 986.9 3 from ($^{14}\text{N}, 3\text{py}$) and 986.6 3 from ($^{18}\text{O}, 2\alpha\text{ny}$). I γ : from ($^{14}\text{N}, 3\text{py}$). Mult.: from ($^{18}\text{O}, 2\alpha\text{ny}$) and ($^{14}\text{N}, 3\text{py}$). B(E2)(W.u.)=0.22 +5-4
		3018.5 7	43 5	1991.27	7/2 ⁻	E2	0.000798 11	Mult.: from $\gamma\gamma(\theta)$ (ADO) and RUL in ($^{14}\text{N}, 3\text{py}$). E γ : from ($^{14}\text{N}, 3\text{py}$). Other: 1389.0 8 from ($^{18}\text{O}, 2\alpha\text{ny}$). I γ : from ($^{14}\text{N}, 3\text{py}$). Mult.: from $\gamma\gamma(\theta)$ (ADO) in ($^{14}\text{N}, 3\text{py}$). E γ , I γ : from ($^{14}\text{N}, 3\text{py}$). E γ , I γ : from ($^{14}\text{N}, 3\text{py}$). E γ , I γ : from (n, γ) E=thermal. E γ , I γ : from (n, γ) E=thermal.
5412.28	(9/2 ⁺)	1389.0 4	100 10	4023.24	(11/2 ⁻)	D		E γ : weighted average of 465.3 3 from ($^{14}\text{N}, 3\text{py}$) and 465.5 2 from ($^{18}\text{O}, 2\alpha\text{ny}$). I γ : unweighted average of 31.6 26 from ($^{14}\text{N}, 3\text{py}$) and 43.8 32 from ($^{18}\text{O}, 2\alpha\text{ny}$). Mult.: from ($^{18}\text{O}, 2\alpha\text{ny}$). E γ : weighted average of 867.3 3 from ($^{14}\text{N}, 3\text{py}$) and 868.2 4 from ($^{18}\text{O}, 2\alpha\text{ny}$). I γ : from ($^{14}\text{N}, 3\text{py}$). Other: 62 7 from ($^{18}\text{O}, 2\alpha\text{ny}$). Mult.: from ($^{18}\text{O}, 2\alpha\text{ny}$). E γ : weighted average of 977.8 3 from ($^{14}\text{N}, 3\text{py}$) and 978.4 10 from ($^{18}\text{O}, 2\alpha\text{ny}$). I γ : weighted average of 42.1 26 from ($^{14}\text{N}, 3\text{py}$) and 82 22 from ($^{18}\text{O}, 2\alpha\text{ny}$). Mult.: from ($^{14}\text{N}, 3\text{py}$). E γ : unweighted average of 1054.5 3 from ($^{14}\text{N}, 3\text{py}$) and 1055.6 4 from ($^{18}\text{O}, 2\alpha\text{ny}$).
		1525.7 8	15.4 26	3886.28	(5/2 ⁺)			
		3420.6 9	28.2 26	1991.27	7/2 ⁻			
5752.5	5/2 ⁻ , 7/2 ⁻	5752.0 8	100	0.0	3/2 ⁺			
5841.482		4268.65 14	100	1572.370	1/2 ⁺			
5877.70	(11/2 ⁺)	465.4 2	38 6	5412.28	(9/2 ⁺)	D		
		867.6 4	50.0 26	5010.05	(11/2 ⁻)	D		
		977.9 3	43 5	4899.82	(9/2 ⁺)	D+Q		
		1055.1 6	80 7	4822.79	(9/2 ⁺)	D		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\alpha^@$
							Comments
							I_γ : weighted average of 79.5 from ($^{14}\text{N}, 3\text{py}$) and 122.32 from ($^{18}\text{O}, 2\alpha\text{ny}$). Mult.: from ($^{18}\text{O}, 2\alpha\text{ny}$). E_γ : from ($^{14}\text{N}, 3\text{py}$). Other: 1854.542 from ($^{18}\text{O}, 2\alpha\text{ny}$). I_γ : from ($^{14}\text{N}, 3\text{py}$). Other: 88.44 from ($^{18}\text{O}, 2\alpha\text{ny}$). E_γ : weighted average of 2061.53 from ($^{14}\text{N}, 3\text{py}$) and 2063.314 from ($^{18}\text{O}, 2\alpha\text{ny}$). I_γ : from ($^{14}\text{N}, 3\text{py}$). Other: 100.20 from ($^{18}\text{O}, 2\alpha\text{ny}$). Mult.: from ($^{18}\text{O}, 2\alpha\text{ny}$). E_γ : weighted average of 2282.94 from ($^{14}\text{N}, 3\text{py}$) and 2283.510 from ($^{18}\text{O}, 2\alpha\text{ny}$). I_γ : unweighted average of 68.426 from ($^{14}\text{N}, 3\text{py}$) and 44.10 from ($^{18}\text{O}, 2\alpha\text{ny}$). E_γ, I_γ : from ($^{14}\text{N}, 3\text{py}$). E_γ, I_γ : from (n, γ) E=thermal. E_γ, I_γ : from (n, γ) E=thermal. E_γ, I_γ : from (n, γ) E=thermal. E_γ, I_γ : from (n, γ) E=thermal.
5877.70	(11/2 ⁺)	1854.4 4	44.7 26	4023.24	(11/2 ⁻)		
		2061.6 4	100 5	3816.01	(9/2 ⁻)	D	
		2283.0 4	56 12	3594.62	(7/2 ⁺)		
6018.8		3886.0 13	7.9 26	1991.27	7/2 ⁻		
6078.475	1/2 ⁻ , 3/2 ⁻	6018.2 6	100	0.0	3/2 ⁺		
		3139.9 5	22 7	2938.62	(3/2 ⁺)		
		6077.87 11	100 12	0.0	3/2 ⁺		
6293.93		6293.2 4	100	0.0	3/2 ⁺		
6299.4	(11/2 ⁺)	887.0 7	40 10	5412.28	(9/2 ⁺)	D	
		2275.9 6	100 20	4023.24	(11/2 ⁻)		
		2483.6 8	90 20	3816.01	(9/2 ⁻)		
6352.2	(13/2 ⁻)	2329.5 9	100 20	4023.24	(11/2 ⁻)	D	
		2535.8 11	50 6	3816.01	(9/2 ⁻)	E2	0.000581 8
6354.89		6355.0 6	100	0.0	3/2 ⁺		
6419.9		6419.3 11	100	0.0	3/2 ⁺		
6629.43		6628.5 6	100	0.0	3/2 ⁺		
6761.0		6760.3 12	100	0.0	3/2 ⁺		
(6986.096)	1/2 ⁺	356.66 & 9	<0.073 &	6629.43			
		631.32 & 24	<0.123 &	6354.89			
		692.16 5	0.238 31	6293.93			
		907.608 23	1.00 18	6078.475	1/2 ⁻ , 3/2 ⁻		
		1144.591 20	0.98 11	5841.482			
		2022.954 9	19.6 18	4963.081	3/2 ⁻		
		2082.83 17	28.4 27	4903.371	1/2 ⁻		
		2508.39 8	0.68 7	4477.63	(5/2 ⁺)		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\alpha^{\text{@}}$	Comments
(6986.096)	1/2 ⁺	2796.74 4	8.8 7	4189.238	1/2 ⁻			
		3183.95 4	10.9 11	3801.952	3/2 ⁻			
		4638.14 14	100 9	2347.779	3/2 ⁻			
		6985.7 10	0.064 14	0.0	3/2 ⁺			
7179.74	(15/2 ⁺)	827.8 6	2.2 8	6352.2	(13/2 ⁻)	[E1]	3.08×10 ⁻⁵ 4	B(E1)(W.u.)=7×10 ⁻⁶ +6-3
		880.2 4	5.9 8	6299.4	(11/2 ⁺)	[E2]	6.66×10 ⁻⁵ 9	B(E2)(W.u.)=2.8 +18-8
		1302.0 3	100 6	5877.70	(11/2 ⁺)	E2	5.56×10 ⁻⁵ 8	B(E2)(W.u.)=7 +4-2 E _γ : weighted average of 1302.2 2 from (¹⁴ N,3pγ) and 1301.6 3 from (¹⁸ O,2αnγ). I _γ : from (¹⁴ N,3pγ). Mult.: from γγ(θ)(ADO) and RUL in (¹⁴ N,3pγ). E _γ : weighted average of 843.9 8 from (¹⁴ N,3pγ) and 844.3 6 from (¹⁸ O,2αnγ). Mult.: from (¹⁸ O,2αnγ).
8024.0	(17/2 ⁺)	844.2 6	100	7179.74	(15/2 ⁺)	D		
8755.9	(19/2)	731.6 15	57 22	8024.0	(17/2 ⁺)	D		
		1576.3 14	100 22	7179.74	(15/2 ⁺)			
10048.6	(19/2)	2868.7 8	100	7179.74	(15/2 ⁺)	Q		
12469.3	(21/2)	2420.6 11	100	10048.6	(19/2)	D		

[†] E_γ values with uncertainties are primarily from ³⁴S(n,γ) (E=thermal), ²⁴Mg(¹⁴N,3pγ) and ²⁶Mg(¹⁸O,2αnγ) fusion-evaporation reactions. E_γ values without uncertainties are deduced from level-energy differences.

[‡] I_γ values are primarily from ³⁴S(n,γ) (E=thermal), ³⁷Cl(d,αγ), ³⁴S(d,pγ), ²⁴Mg(¹⁴N,3pγ) and ²⁶Mg(¹⁸O,2αnγ) fusion-evaporation reactions.

[#] Mult and δ are primarily from γ(θ), γγ(θ), and γγ(θ)(ADO) in ³⁴S(d,pγ), ²⁴Mg(¹⁴N,3pγ), and ²⁶Mg(¹⁸O,2αnγ). The magnetic or electric nature is determined based on RUL and measured T_{1/2} where available.

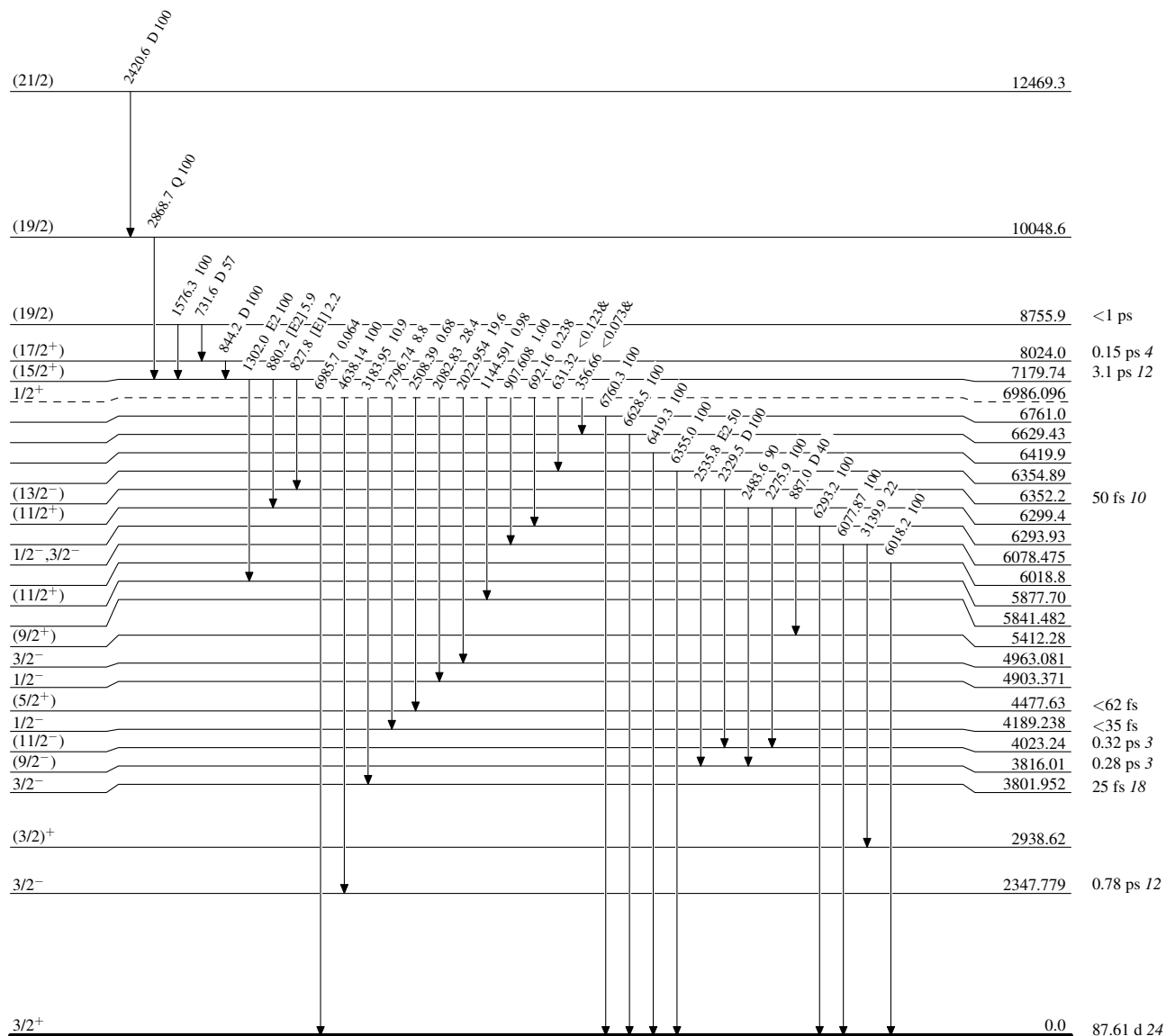
[@] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

[&] Multiply placed with undivided intensity.

^a Placement of transition in the level scheme is uncertain.

Adopted Levels, GammasLevel Scheme

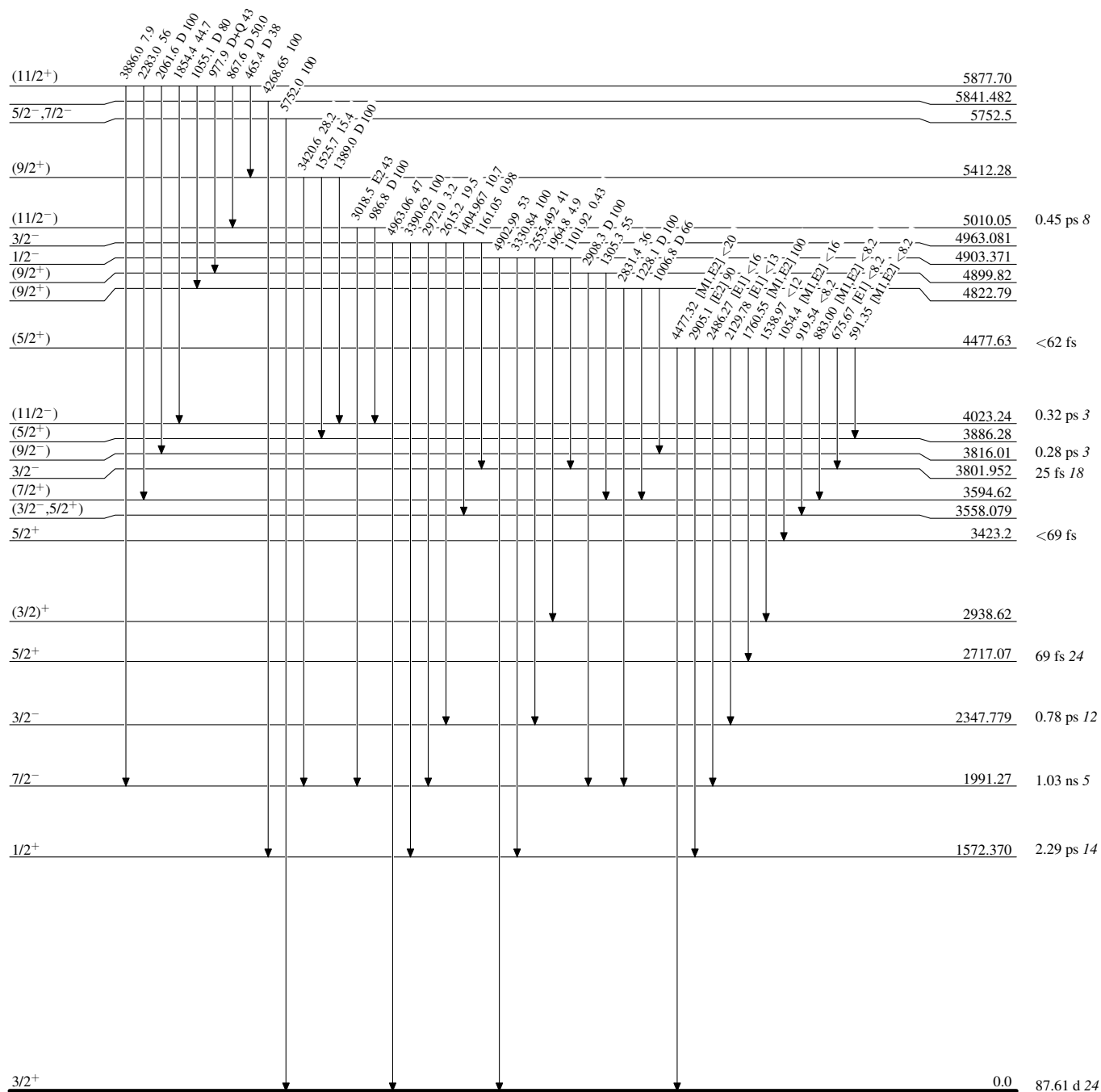
Intensities: Relative photon branching from each level
& Multiplied placed: undivided intensity given

 $^{35}_{16}\text{S}_{19}$

Adopted Levels, Gammas

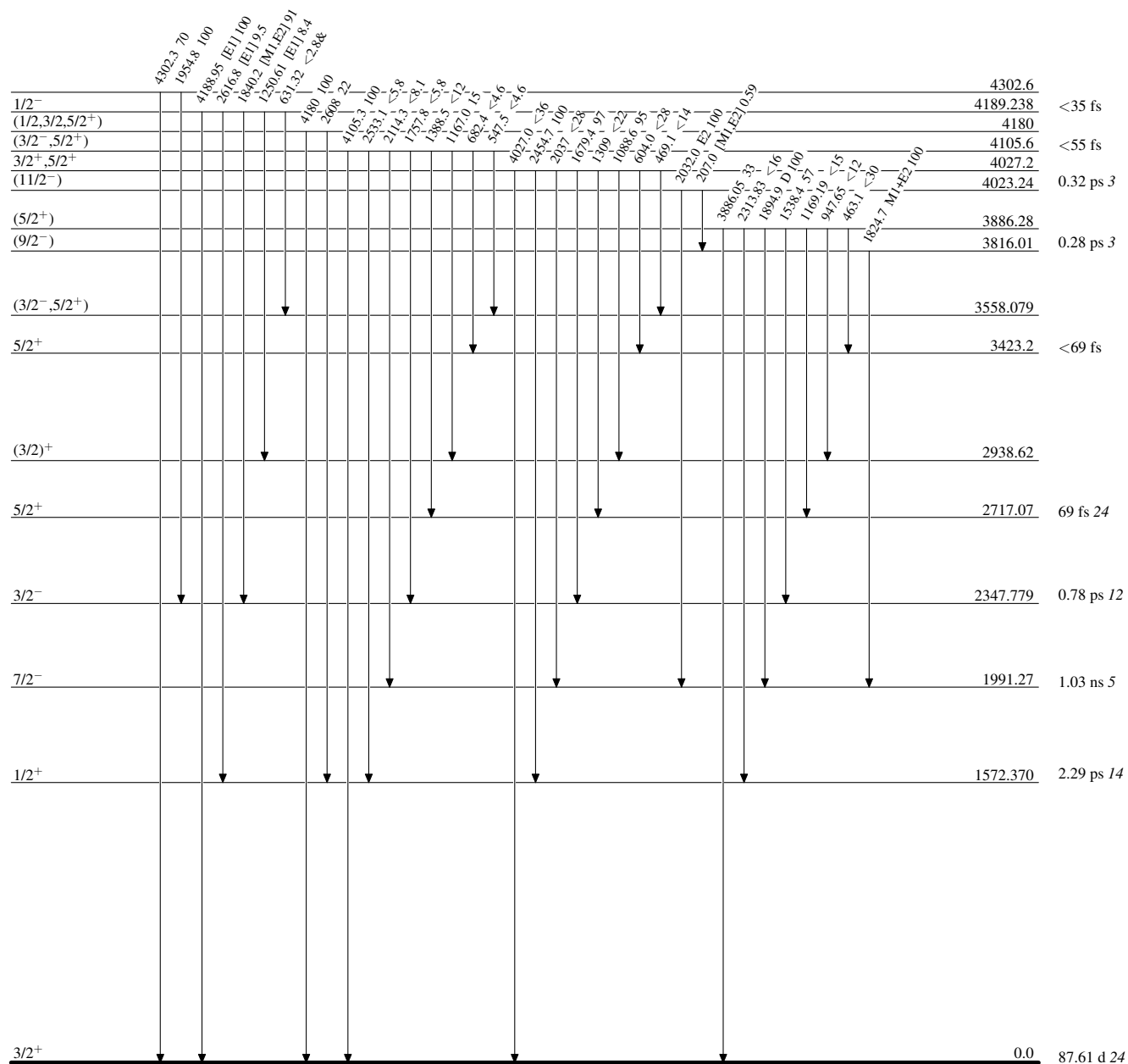
Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

-----▶ γ Decay (Uncertain)

